



Luis F. Escobar, Ph.D. Candidate

## Contact Information

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## Research Interests

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Robotics, GNSS-denied Autonomous Systems, ROS 2, Multi-Robot Coordination, UAV Path Planning, UAV Energy Optimization, Industrial Automation, Mechatronics Design, and Engineering Education.

## Education

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**West Virginia University** 2023-May 2026 (Expected)

*PhD Candidate in Robotics*

Morgantown, USA

*Dissertation Topic: "Energy Aware path planning for multi rotor UAVs"*

**University of Pennsylvania** 2021-2022

*PhD Student (Transferred)*

Philadelphia, USA

**Waseda University** 2012-2015

*M. Eng. in Information, Production and Systems Engineering*

Kitakyushu, Japan

**ITESM - Monterrey Institute of Technology** 2004-2009

*B. Sc. Mechatronics Engineering*

Monterrey, Mexico

## Research & Professional Experience

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**West Virginia University** 2023-Present

*Graduate Research Assistant*

Morgantown, USA

*Developing autonomous navigation stacks for GNSS-denied underground environments. Leading the complete mechatronic design and prototyping of multirotor UAVs. Implementing energy-aware optimization strategies for coverage path planning.*

**Universidad de las Fuerzas Armadas ESPE** 2015-2021

*Professor & Head of Laboratory*

Sangolqui, Ecuador

*Directed the Robotics Laboratory and led the "Best Research Team" (2019). Supervised undergraduate theses in industrial automation and mechatronics design.*

**Lainbori S.A.** 2016-2021

*Senior Engineer*

Quito, Ecuador

*Led continuous improvement initiatives for fiber optic maintenance teams. Implemented Kanban systems and Project Management strategies to optimize operational workflows.*

**Universidad de las Fuerzas Armadas ESPE**

2009-2012

*Mechatronics Program Professor*

Sangolqui, Ecuador

*Led the implementation of the undergraduate Mechatronics Engineering program. Coordinated curricula across Mechanics, Electronics, and IT departments. Served as the lead instructor for capstone courses as the university's sole Mechatronics Engineer.*

**Ternium**

2009

*Continuous Improvement Intern*

Monterrey, Mexico

*Collaborated with Six Sigma Black Belts to execute cost-reduction and safety projects. Supported manufacturing leaders in the implementation of Six Sigma tools.*

**Volkswagen AG**

2008

*Production Systems Intern*

Kassel, Germany

*Analyzed automatic transmission gear manufacturing within the Volkswagen Production System. Conducted time and motion studies to optimize operator ergonomics and reduce CNC setup times.*

## **Publications**

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**2026**

- "Development of an Autonomous UAV for Multi-Modal Mapping of Underground Mines", *Localization and 3D Mapping of Intelligent Robotics. Robotics*.

DOI: 10.3390/robotics15030063

**2025**

- "Energy-Aware Coverage Path Planner for Multirotor UAVS", *IEEE ICUAS 2025*.

DOI: 10.1109/ICUAS65942.2025.11007830

**2022**

- "Implementation of Collaborative Work Between Two SCARA Robots in a Robotic Cell for Continuous Classification of Products", *Lecture Notes in Electrical Engineering*.

DOI: 10.1007/978-3-031-08288-7<sub>7</sub>

- "Plant Layout Selection Procedure Based on Discrete Event Simulation Software", *Lecture Notes in Electrical Engineering*.

DOI: 10.1007/978-3-031-08280-1<sub>4</sub>

- "Process Optimization with Discrete Event Simulation Software: An Experience in Ecuador Small Enterprise", *Lecture Notes in Electrical Engineering*.

DOI: 10.1007/978-3-031-08280-1<sub>5</sub>

- "Design, Construction and Evaluation of a 3DOF Urban Planting System", *Lecture Notes in Electrical Engineering*.

DOI: 10.1007/978-3-031-08288-7<sub>6</sub>

- "Multi-robot System for Collaborative Work Equipped with Trajectory Planning over IoT Architecture", *Lecture Notes in Networks and Systems*.

DOI: 10.1007/978-3-030-90033-5<sub>24</sub>**2020**

- "Flexible Manufacturing Systems Optimization with Meta-heuristic Algorithm Using Open Source Software", *Lecture Notes in Electrical Engineering (CIT 2020)*.

DOI: 10.1007/978-3-030-72212-8<sub>18</sub>

- "Multi-Robot platform with features of Cyber-physical systems for education applications", *IEEE ANDESCON 2020*.

DOI: 10.1109/ANDESCON50619.2020.9272030

- "Validation through a digital twin of a Stewart platform with irregular geometry with 6 DOF for simulation of a transport vehicle", *IEEE CASE 2020*.

DOI: 10.1109/CASE48305.2020.9216995

- "YaniWawa: An Innovative Tool for Teaching Using Programmable Models over Augmented Reality Sandbox", *Advances in Intelligent Systems and Computing*.

DOI: 10.1007/978-3-030-59194-6<sub>31</sub>

- "6 DOF anthropomorphic robot as a platform for teaching robotics", *2020 IEEE/ASME AIM*.

DOI: 10.1109/AIM43001.2020.9158828

- "Construction of a Computer Vision Test Platform: VISART for Facial Recognition in Social Robotics", *Communications in Computer and Information Science*.

DOI: 10.1007/978-3-030-42520-3\_50

- "Development of a Social Robot NAR for Children's Education", *Advances in Intelligent Systems and Computing*.

DOI: 10.1007/978-3-030-32033-1\_33

### 2019 and Prior

- "Design of an Autonomous Mobile Robot as a Base Platform for Research of Cyber Physical Systems", *ICAT 2019*.

DOI: 10.1007/978-3-030-42517-3\_16

- "Mechatronics over time: 12 years of creative experiences in Ecuador", *LACCEI 2019*.

DOI: 10.18687/LACCEI2019.1.1.451

- "A New Real-Time Flight Simulator for Military Training Using Mechatronics and Cyber-Physical System Methods", *InTechOpen*, 2019.

DOI: 10.5772/intechopen.86586

- "Implementation of an IoT Architecture based on MQTT for a Multi-Robot System", *IEEE ETCM*, 2018.

DOI: 10.1109/ETCM.2018.8580321

- "Design of a Spatial Disorientation Simulator using a Modified Stewart-Gough Platform", *IEEE ETCM*, 2018.

DOI: 10.1109/ETCM.2018.8580347

- "Desarrollo de un simulador de sistemas de manufactura flexible con interfaz gráfica basado en redes de petri", *Revista Iberoamericana de Ingeniería Mecánica*, 2018.

- "Real-time flight simulator construction with a network for training pilots using mechatronics and cyber-physical system approaches", *IEEE ICPCSI*, 2017.

DOI: 10.1109/ICPCSI.2017.8392169

- "Design and implementation of complex systems using Mechatronics and Cyber-Physical Systems approaches", *IEEE ICMA*, 2017.

DOI: 10.1109/ICMA.2017.8015804

- "Kinematic resolution of delta robot using four bar mechanism theory", *IEEE ICMA*, 2017.

DOI: 10.1109/ICMA.2017.8015932

## Technical Skills

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- **Robotics Middleware:** ROS 2 (Navigation, Simulation, Deployment), MoCap systems.
- **Design & Prototyping:** SolidWorks, Fusion360, Catia, 3D Printing, Mechanical Analysis (CAD/CAM/CAE).
- **Embedded Systems:** Microcontrollers (ESP32, Arduino, PIC), PCB Design, Signal Processing.
- **Robotic Manipulators:** KUKA, Fanuc, Motoman, Franka Emika.
- **Programming:** Fluent in Python and C++ (Git, Linux/Ubuntu), MATLAB, LabView.

## Honors & Awards

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- **Fulbright Scholarship** to pursue a PhD degree in the USA (2021).
- **Monbukagakusho (MEXT) Scholarship**, Japan (2013).
- **DAAD Scholarship**, Germany (2007).
- Indian Government Scholarship for Mechatronics and its Application Program (2011).
- **Best Research Team** for Projects of National Interest, ESPE University (2019).
- **First Place:** IEEE Best Technical Chapter in Latin America (RAS) (2018, 2019).

## Professional Service & Leadership

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### IEEE Robotics & Automation Society (RAS) - Ecuador Section

- President (2020)
- Vice-President (2019)
- Student Branch Counselor (2010-2012 and 2016-2021)

### Invited Speaker & Organizer

- Delivered multiple technical seminars and workshops on Robotics and Mechatronics at various universities and government institutions across Ecuador (2015-2021), including events for the Ministry of Education and universities with active IEEE RAS student branches.

## References

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### Dr. Guilherme A. S. Pereira

Professor

Department of Mechanical, Materials and Aerospace Engineering

West Virginia University

Email: guilherme.pereira@mail.wvu.edu

### Dr. Jason Gross

Chair and Professor

Department of Mechanical, Materials and Aerospace Engineering

West Virginia University

Email: jason.gross@mail.wvu.edu

### Dr. Luis Segura

Assistant Professor

Industrial & Systems Engineering

University of Louisville

Email: ljsegu01@louisville.edu